

FULL ARTICLE



Political competition, spatial interactions, and default risk of local government debts in China

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Abstract

The paper investigates the institutional mechanism behind increasing local government debts and default risk in the last decade in China. Using prefectural-level data from 2010 to 2017, we study the default risk on local government debts through the lens of spatial interactions. Empirical results identify positive spatial interactions of default risk among political neighbouring cities, but not for merely geographic or economic neighbours. Moreover, the spatial interactions vary with local leader's political incentives. These findings altogether suggest the spatial interactions of default risk is driven by political competition among local leaders. The study also rules out alternative explanations such as resource flow effects.

KEYWORDS

default risk, local government debts, political competition, spatial interactions

JEL CLASSIFICATION

R14, H73, P16

1 | INTRODUCTION

Due to the enactment of fiscal and monetary stimulus plan to counteract economic recession, many countries such as the US, Japan and Spain have experienced ever-increasing public debt after the 2008 Global Financial Crisis. At the same time, the large and growing government debts at sub-national levels are gaining attention from policy-makers and citizens, especially after the Detroit bankruptcy in 2013. According to Eurostat, the declining public revenue together with a collapsing economy led to mounting government debts. The debt-to-GDP ratio of the European Union rose to 86.7% in 2017, and the main countries stood at 64.1% for Germany, 97% for France and 131.8% for



Italy. As an emerging economy, China also experienced massive local government debts in the last decade. The explicit debt and implicit debt of Chinese local governments amounted to 16.47 trillion and 15.42 trillion respectively in 2017, and the debt-to-GDP ratio of Guizhou Province and Qinghai Province both have surpassed 100%.

Previous studies have explored the determinants of local government debts. The relevant contributions include, among others, Ashworth et al.'s (2005) study of Flemish municipalities, Baber and Gore's (2008) analysis of US cases, and Benito and Bastida (2004) and Balaguer-Coll et al.'s (2016) focus on local debts in Spain. Based on these studies, the factors influencing local debt accumulation could be summarized into three types: fiscal, socio-economic and political factors. However, compared with democratic society, the political factors, especially political competition play a more crucial role in Chinese local debt accumulation, under the top-down political power structure in which higher level officials hold the power to promote their lower level counterparts (Bai et al., 2016; Qu et al., 2019; Xiong, 2018).

However, the spatial interactions of debt default risk in the context of political competition, which threatens financial stability and sustainable development, have hitherto received scant attention. Actually, Chinese local officials' political payoff is determined by Relative Performance Evaluation (RPE) compared with their counterparts, the political competition fits within the information spillover model in the sense that local officials strategically respond to policy changes in other jurisdictions or adopt mimetic behaviour to get political promotion (Brueckner, 2003; Li & Zhou, 2005; Yao & Zhang, 2015). With existing literatures identifying interactions among local governments in public expenditure, public services provision, etc., in the political competition framework, the overarching aim of our study is to explore the spatial interactions of debt default risk.

With a dataset on Urban Construction Investment Bonds (UCIBs, *chengtouzhai*) from 2010 to 2017, we derive the default risk at prefectural level, and explore its spatial interactions. First, we calculate the debt default risk, and describe its temporal and spatial characteristics. The default risk remained stable between 2010 and 2015 but rose abruptly after 2015. And the default risk of prefectural-level cities in northern regions, especially Inner Mongolia and three northeast provinces, accumulates relatively faster compared to other cities. Then we lay out a simple model to discuss the mechanism behind spatial interactions, and employ the Spatial Panel Data Model (SPDM) to explore spatial interactions of default risk on local government debts. With five spatial weight matrices based on political, economic and geographical proximity, we identify remarkable spatial interactions among political neighbours rather than other types of neighbours, implying that the interactions stem from political competition. We also observe stronger spatial interactions among cities whose party secretaries are at crucial points in their career, indicating positive effects of political incentives in regional strategic interactions. In addition, heterogeneity analysis indicates that central-western cities have experienced stronger spatial interactions than eastern cities. Finally, we examine the impacts of policy change on local financing, and find that it alleviates spatial interactions to some extent by allowing direct and transparent financing vehicles, but does not fully eliminate them.

People may argue that there are still other possible interpretations for spatial interactions of default risk besides political competition, for example, the economic spillover and resource flow. The economic spillover hypothesis argues that the economic development of surrounding regions has a positive impact on local fiscal or economic growth, while the resource flow hypothesis shows that local leaders always provide pro-business policies to attract capital and labour from other regions (Besley & Case, 1995; Brueckner, 2003; Tiebout, 1956). Given that resource flows or economic spillovers heavily rely on the geographic distance, we construct two spatial weight matrices based on geographical proximity among cities to address these concerns. The empirical results show that there are no spatial interactions among geographically neighbouring cities. Thus, the combined results suggest that spatial interactions of default risk are induced by political competition instead of economic spillover or resource flow among cities.

Our study is related to three strands of literature. The first one focuses on local government debts and its risk. Previous studies mainly focus on the evolution, importance, geographic distribution and underlying financial mechanisms of local government debts (Bai et al., 2016; Feng et al., 2022; Tao, 2015). According to Cuestas and Regis (2018), Chinese local debt and its risk have witnessed rapid growth after announcement of post-crisis stimulus programmes. The rising trend of local debts in other countries has also been revealed (Fournier & Fall, 2017;



Yared, 2019). Xiong (2018) and Qu et al. (2019) attribute the surge of Chinese local debts to political competition, in which local officials increase leverage borrowings to spur total investment and economic growth competitively. Buiatti et al. (2014) claim that Italy's debt accumulation is driven by persistent fiscal imbalance in southern areas. Whereas most existing studies have discussed debt accumulation and its risk in many counties, our unique contribution to this strand of literature is to calculate Chinese debt default risk at prefectural level and illustrate its spatial and temporal dynamics.

The second related strand of literature discusses spatial interactions among local governments with two main mechanisms: spillover effects and resource flow (Belenzon & Schankerman, 2013; Brueckner, 2003; Tiebout, 1956). Much existing literature focuses on strategic interactions of fiscal policy-making, such as those concerning fiscal expenditures and tax revenue. (Bocci et al., 2019; Huang & Du, 2017; Li & Xiong, 2019; Willardsen, 2021; Yu et al., 2013). With respect to the spatial interactions of local government debts, Pan et al. (2017) argue that inter-jurisdictional political competition promotes local leaders to issue UCIBs and finance economic development competitively, leading to spatial interactions of local debt accumulation. Baskaran (2012) claims the German state reacts to debt decisions of other states due to soft budget constraint. Another study from Borck et al. (2015) explores German spatial interactions of municipalities' debt in Bavaria and North Rhine-Westphalia separately. Similarly, Balaguer-Coll et al. (2016) has also proved the presence of significant spatial effects of local government debts in Valencia (Spain). Based on the co-movement of government expenditure, revenue and debt, we further explore the spatial interactions of debt default risk among prefectural-level cities.

Finally, our paper speaks to a large literature on political competition between local officials. As summarized by Xu (2011), two key features of China's fundamental institution presumably inspired local officials: the political centralization and economic regional decentralization. The former enables central government to maintain personal control of local leaders, while the later gives local governments substantial discretionary power to intervene in economic development, such as providing public infrastructure, allocating financial resources and even making policy innovations within their jurisdictions. In this institutional context, local leaders compete with their rivals through devoting to objectives of central government, which is called the pattern of 'performance-based tournament competition' (Chen et al., 2005; Choi, 2012; Li & Zhou, 2005). Comparing with yardstick competition in western countries, the tournament competition is based on top-down power structure in which regional officials are appointed by up-level authority, instead of being elected by local electorate (Besley & Case, 1995; Xiong, 2018; Yu et al., 2016). With the Chinese government's overriding task shifting to economic growth in the post-Mao era, the political promotion propels local leaders to be keen on the projects leading to fast economic growth (e.g., real estate and infrastructure construction) and then deliver satisfactory performance and impress their superiors. However, the overemphasis on economic growth distorts the structure of local governments' expenditure, and then results in excessive borrowings in public sector, which jeopardizes sustainability of economic development (Cuestas & Regis, 2018; Liu et al., 2020). Our paper analyzes the relevant mechanism responsible for spatial interactions of debt default risk, and then sheds light on the impacts of political competition on risk management.

The rest of paper is organized as follows. Section 2 introduces the background, development and geographic distribution of UCIBs. Section 3 provides the theoretical framework to discuss mechanisms of spatial interactions. Empirical specifications and data are in Section 4. Section 5 discusses the empirical results. Section 6 concludes briefly with policy implications.

2 | UCIBS IN CHINA

2.1 | A brief introduction of UCIBs

The tax-sharing reform (*fenshuizhi*) in 1994, aimed at recentralizing fiscal power and decreasing the proportion of local government revenue in the total revenue from 77.98% in 1993 to 52.99% in 2017 (Che et al., 2017; Van der



Kamp et al., 2017). At the same time, local governments are still responsible for most infrastructure construction and public services, leading to huge fiscal pressure. Due to prohibition from direct market borrowing according to Budget Law, local governments started to establish local government financing platforms (LGFPs, *defangzhengfurongzippingtai*) and injected state-owned assets and government subsidies into them, in order to issue UCIBs to finance local infrastructure projects. However, due to little policy support from central government, the issued scale of UCIBs in this period remained comparably small and most UCIBs concentrated on economically developed provinces such as Jiangsu and Zhejiang.

Confronting the global financial crisis in 2008, the Chinese central government enacted a 4 trillion-RMB stimulus plan to cope with slowdown in economic growth, which prioritized infrastructure projects such as railways, airports, and water conservancy. However, the central government only offered 1.8 trillion-RMB for this plan, and local governments were responsible for the remaining 2.2 trillion-RMB. In order to meet local governments' financing demand, the People's Bank of China (PBC) officially specified UCIBs as the financial vehicle in 2009 and encouraged local governments to finance infrastructural investment via these vehicles. As a result, local governments issued UCIBs immoderately to foster local economic growth, resulting in rapid growth in the amount of UCIBs' issued. In 2013, local governments were responsible for nearly 7 trillion-RMB contingent liabilities, and almost all of them were UCIBs. The over-issue of UCIBs indicates that the main goal of local governments is boosting economic growth through financing infrastructural investment instead of merely filling the fiscal gap. With increased financing reliance on UCIBs, the LGFPs have extensively engaged in many sorts of welfare projects enacted by local governments.

Although the 4 trillion-RMB stimulus plan prompts urban infrastructural development and rescues China from economic recession, it also resulted in massive local government debts and consequent debt default risk (Fan & Lv, 2012). In order to strengthen the management of local government debts, central government revised the Budget Law in 2014, and opened an outlet for local governments to direct market borrowings. The new Budget Law empowers provincial governments to issue local government bonds on behalf of the lower-level governments (e.g., prefectural-level and county-level governments) after 2015. However, the issued amount of local government bonds still had to be approved by central government. Therefore, due to little autonomy in bonds issuance, local governments still regard UCIBs as important financing vehicles (Feng et al., 2022), thus the issued amount of UCIBs continued to surge after 2015. The discrepancy between central government's policy preference and local behaviours could be driven by the political competition among local leaders. The economic performance determines local leaders' chance of political promotion, and thus prompts them to finance local development with more flexible vehicles like UCIBs.

2.2 | The development of UCIBs between 2009 and 2017

The fiscal pressure of local governments, further combined with the 4 trillion-RMB economic stimulus plan, leads to massive expansion of UCIBs after 2008. We summarize the issued amount and outstanding amount of UCIBs at prefectural level in Figure 1. First, the annually issued amount of UCIBs climbed continuously from 2009, and reached its peak of 2.5 trillion RMB in 2016. Because of tight refinancing policy publicized at the end of 2016, the number dropped to 1.97 trillion in 2017. At the same time, local governments had rising repayment pressure due to aggressive expansion during this period. By the end of 2017, outstanding UCIBs amounted to 7.94 trillion RMB, while the annual fiscal revenue of local governments remained at only 9.14 trillion according to Chinese Ministry of Finance.

Table 1 summarizes the credit ratings of outstanding UCIBs by 2017. Only 4.13% of UCIBs' credit ratings are evaluated as AA- or below, which seems at odds with its high default risk reported in Bo et al. (2017). The reason might be that UCIBs' credit ratings only reflect the overall risk, instead of individual default risk by local governments due to soft budget constraints. On the other hand, credit rating shopping might be another reason for overestimated credit ratings (Sangiorgi & Spatt, 2016). In order to compete for more business, credit-rating agencies have an incentive to grant bond issuers higher credit ratings than they deserve.

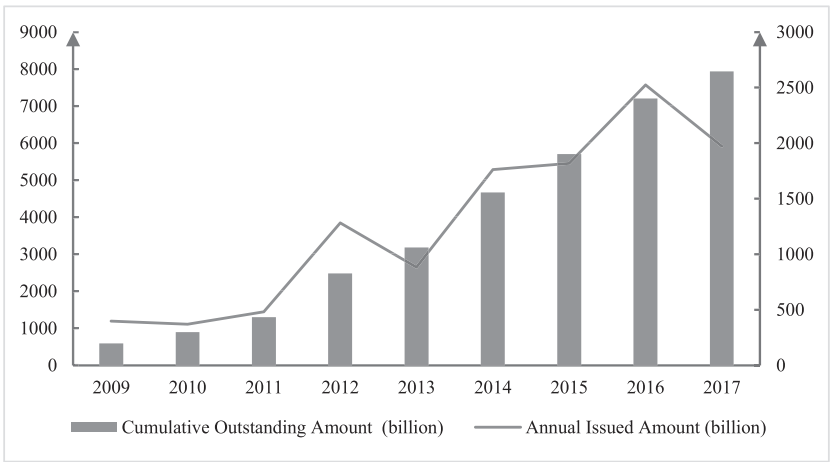


FIGURE 1 Annually issued amount and outstanding amount of UCIBs during 2009 and 2017

TABLE 1 Outstanding UCIBs by credit rating

Credit rating	Number	Percentage
AA- and below	222	4.13
AA	2,780	51.76
AA+	1,578	29.38
AAA	791	14.73
Obs.	5,371	100

Figure 2 describes the quarterly pattern of UCIBs' coupon rate across the period of 2009–2017. Above all, the UCIBs' coupon rate stayed significantly lower than that of corporate bonds in the same period because of local governments' implicit guarantee. But the UCIBs' coupon rate is relatively higher than treasury bonds due to the absence of the central government's explicit guarantee for repayment. Moreover, UCIBs' coupon rate fluctuated with government policy and economic conditions. It stayed at a fairly high level between 2009 and 2014 due to 4 trillion-RMB stimulus package. To combat economic downturn, UCIBs experienced a relatively loose issuance policy in 2015 and the first half of 2016, thus their coupon rate declined during this period. But after September 2016, UCIBs held a rising financing cost with high coupon rate because of strict bail-out policy.

Figure 3 depicts geographic distribution of UCIBs issued between 2009 and 2017, where prefectural cities fall into four groups according to their quartiles in issued amount. Cities differ largely in their issued amount of UCIBs, with more than half of the cities below 20 billion RMB, and only a tiny portion over 100 billion RMB. Obviously, the issued amount of UCIBs seems to correlate with the economic advancement across regions, where the relatively rich eastern and central regions claim more bonds than the western regions.

3 | THEORETICAL FRAMEWORK

The paper's main goal is to explore the spatial interactions of default risk on local government debts. In this section, we draw on theoretical analysis in Brueckner (2003) and Shi and Xi (2016), and lay out a simple model to understand the spatial interactions of debt default risk.

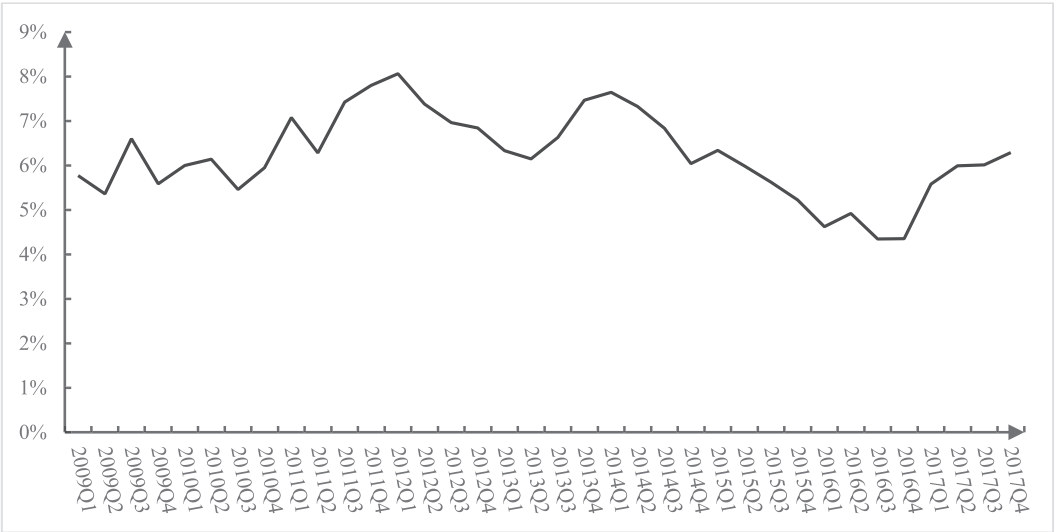


FIGURE 2 Change of UCIB's coupon rate

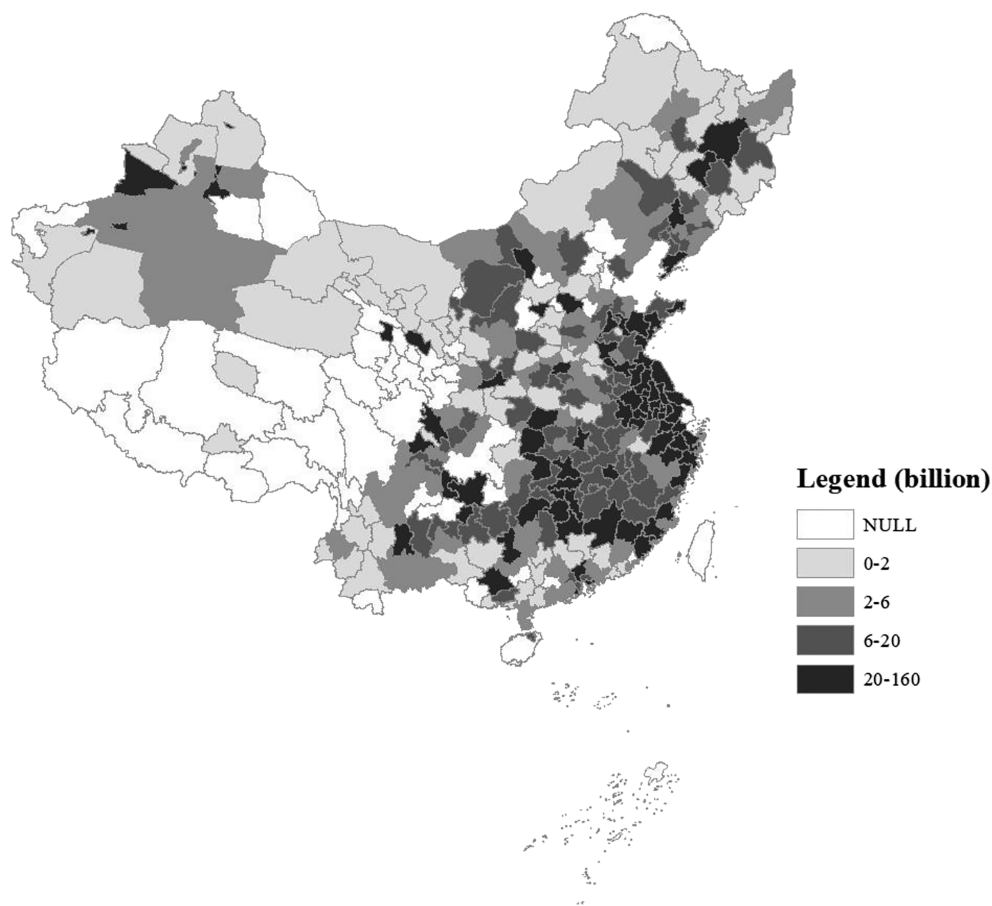


FIGURE 3 Geographic distribution of UCIBs issued during 2009 and 2017



In China's top-down power structure, the higher-level authorities hold the power to promote or dismiss subordinates based on the RPE, which means local officials' career prospects, for example, political promotion, turnover or demotion, is determined by relative performance compared to their political rivals. Thus, local leaders tend to make policy decisions according to the choices of their counterparts, that is, information spillovers from other jurisdictions. Considering N prefectural-level cities denoted by $1, \dots, N$, each local leader decides the level of default risk in his/her jurisdictions, r_i , while the payoff is jointly determined by default risk of their own and other cities based on RPE. And we assume local leaders' performance includes two parts: the economic growth and risk reduction.

Local governments rely heavily on UCIBs to finance government-led investment and economic development in the pattern of debt-driven economy (Pan et al., 2017; Yu et al., 2016). Thus the RPE promotes local leaders to increase UCIBs' issuance strategically. Intuitively, local leaders tended to issue more UCIBs to leverage investment after observing the upsurge of UCIBs' issuance in their political rivals' jurisdictions. It could lead to strategic interactions of debt default risk, especially for cities sharing similar economic potential and repayment capacity. We specify city leader's relative performance in economic growth as follows:

$$E_i = y(r_i, \bar{r}_{-i}; \eta_i), \quad (1)$$

where $\bar{r}_{-i} = \sum_{j \neq i} w_{ij} r_{jt}$ is the weighted average default risk of other cities, and η_i represents city i 's fixed characteristics, such as culture and resource endowment. For given economic and fiscal conditions, loose risk management allows local governments to increase UCIBs' issuance and finance infrastructural investment, leading to better performance in economic growth. Thus, we have:

$$y_1 = \frac{\partial y}{\partial r_i} > 0.$$

Due to decreasing marginal return of government investment, we have:

$$y_{11} = \frac{\partial^2 y}{\partial r_i^2} < 0.$$

According to Brun et al. (2002) and Ouyang and Fu (2012), the city i 's government investment might be more profitable with an increased investment of other cities due to spillover effects, hence:

$$y_{12} = \frac{\partial y}{\partial r_i \partial \bar{r}_{-i}} > 0.$$

Besides economic growth, Chinese central government considers other indicators in local officials' performance evaluation, such as risk reduction, to maintain sustainability of economic development (Chen et al., 2017; Deng et al., 2012). Similar to economic growth, local leaders have to divert efforts to risk reduction according to their counterparts' behaviours, contributing to spatial interactions of debt default risk. Intuitively, reducing debt default risk becomes more urgent for local leaders if their political rivals make substantial progress in this aspect. We specify city leader's relative performance in risk reduction as follows:

$$P_i = g(r_i, \bar{r}_{-i}; \eta_i). \quad (2)$$

Because higher level of r_i indicates worse performance in risk reduction, we have $g_1 = \frac{\partial g}{\partial r_i} < 0$. And we assume non-decreasing marginal revenue, namely, $g_{11} = \frac{\partial^2 g}{\partial r_i^2} \leq 0$. Because local leaders' promotion is decided by the relative



performance in comparison with political rivals, hence $g_{12} = \frac{\partial g}{\partial r_i \partial \bar{r}_{-i}} > 0$. That is, an increase of political competitors' debt default risk would improve local leaders' relative performance in risk reduction, thus leading to larger chance of political promotion.

Suppose local leaders' chance of promotion is decided by the weighted average of performance in economic growth and risk reduction:

$$C_i = \alpha y(r_i, \bar{r}_{-i}; \eta_i) + (1 - \alpha)g(r_i, \bar{r}_{-i}; \eta_i), \quad (3)$$

where α is the weight of regional economic growth in RPE, which is exogenously assigned by the upper-level authority. As the local leaders struggle to maximize the chance of political promotion, we have the first order condition as follows:

$$\alpha y_1 + (1 - \alpha)g_1 = 0. \quad (4)$$

With the implicit function theorem, we have:

$$\frac{dr_i^*}{d\bar{r}_{-i}} = \frac{(\alpha - 1)g_{12} - \alpha y_{12}}{\alpha y_{11} + (1 - \alpha)g_{11}}. \quad (5)$$

Due to $y_{11} < 0$, $g_{11} < 0$, $y_{12} > 0$, $g_{12} > 0$, hence the slope of city i 's response is $\frac{dr_i^*}{d\bar{r}_{-i}} > 0$, indicating positive spatial interactions. The RPE aligns local officials' political interest with relative performance in economic growth and risk reduction. It promotes local leaders to manage debt default risk based on the information spillovers from their political rivals, thus leading to positive spatial interactions of default risk among prefectural-level cities.

4 | RESEARCH DESIGN

In this section, we first discuss the method to calculate default risk on local government debts, and then propose the specifications and spatial weight matrices to capture spatial interactions of default risk among prefectural-level cities, followed by description of the data.

4.1 | Measuring the default risk

We calculate each UCIB's quarterly default risk, and then average over all UCIBs issued by city i in year t , as a proxy of debt default risk. Due to an implicit credit guarantee by central government, UCIBs' credit ratings only reflect the overall risk and cannot serve as an appropriate proxy for default risk. Therefore, we collect UCIBs' daily transaction data, and adopt the method in Dick-Nielsen et al. (2012) and Amihud (2002) to calculate UCIBs' default risk with the following steps.

First, we evaluate the UCIB i 's liquidity risk premium by estimating the regression function below using quarterly trading data:

$$yield_spread_{iq} = \alpha_{iq} + \beta_{iq}liquidity_{iq} + Controls_{iq} + v_q + \varepsilon_{iq}, \quad (6)$$

where $yield_spread_{iq}$ represents difference between UCIB i 's yield and risk-free rate in the q th quarter. $Liquidity_{iq}$ measures UCIB i 's liquidity in the q th quarter. $Controls_{iq}$ represents a set of control variables to mitigate potential



biases caused by omitted variables, including UCIB's credit rating, issued amount, coupon rate and time to maturity. The time effect v_q captures unobservable time-varying factors, such as macro policy effects.

We use three measures for $Liquidity_{iq}$: *Amihud*, *High-low* price spread, and *Turnover*. According to Amihud (2002), we define Amihud as:

$$Amihud_{iq} = \frac{1}{N_{iq}} \left[\sum_t (|R_{itq}| / Volume_{itq}) \right], \quad (7)$$

where N_{iq} represents UCIB i 's total trading days in the q th quarter, R_{itq} is the return of UCIB i on day t in the q th quarter and $Volume_{itq}$ is the respective daily trading volume in RMB. $|R_{itq}| / Volume_{itq}$ gives the absolute price change per RMB standardized by daily trading volume, and $Amihud_{iq}$ captures average impact of order flow on UCIB i 's daily price in the q th quarter. Intuitively, for lower liquid bonds with less trading days and smaller trading volume, one-unit trading has a larger impact on volatility in its return. Therefore, a larger value of $Amihud_{iq}$ means higher liquidity risk. Because high volatility of UCIBs' price indicates illiquidity, the difference between UCIB i 's quarterly highest price and lowest price (*High-low* hereafter) can be used to characterize liquidity risk, with a larger value of *High-low* implying higher liquidity risk. The third measure of UCIBs' liquidity risk, *Turnover*, is defined as:

$$Turnover_{iq} = \frac{Volume_{iq}}{Amount_i}, \quad (8)$$

where $Volume_{iq}$ represents UCIB i 's total trading volume in the q th quarter, and $Amount_i$ is total issued amount. Higher turnover rate suggests the bond is actively transacted, implying lower liquidity risk.

Taking 0.1percentile of *Amihud* and *High-low* and 99.9 quantile of *Turnover* as reference of extremely liquid ($liquidity_{0.1p}$), we acquire UCIB i 's liquidity risk premium in the q th quarter as follows:

$$\eta_{iq} = \beta_{iq} (liquidity_{iq} - liquidity_{0.1p}). \quad (9)$$

The UCIB i 's default risk premium is then retrieved as follows:

$$Default_{iq} = y_{iq} - r_q - \eta_{iq}, \quad (10)$$

where r_q is the q th quarter's risk-free rate with Chinese treasury rate as proxy, y_{iq} and $Default_{iq}$ represent UCIB i 's yield and default risk premium in the q th quarter, respectively. Using the three liquidity indices of *Amihud*, *High_low*, and *Turnover*, we obtain three measures for default risk premium, denoted by $Default_Amihud_{iq}$, $Default_Highlow_{iq}$ and $Default_Turnover_{iq}$. The corresponding measures at a city-year level ($Default_Amihud_{it}$, $Default_Highlow_{it}$ and $Default_Turnover_{it}$) are calculated by averaging over all UCIBs issued by city i in year t , as proxy for default risk on local government debts.

For robustness, we further measure debt default risk from the perspective of repayment ability of local governments. The $Default1_{it}$ is calculated as the ratio of amount of city i 's matured UCIBs to public revenue in year t , and $Default2_{it}$ is defined as the ratio of city i 's issued amount of UCIBs to public revenue in year t .

4.2 | Specifications for the spatial interactions

We adopt the spatial panel data model (SPDM), a flexible model specification approach to capture possible spatial interactions of default risk on local government debts. With extensive availability of panel datasets, the SPDM has



been widely applied in empirics of regional science (Almeida et al., 2021; Donfouet et al., 2018; Fingleton & Arbia, 2008; Hortas-Rico & Rios, 2020; Santolini, 2020). The general expression of SPDM is specified as follows:

$$Risk_{it} = \rho \sum_{j \neq i} w_{ij} Risk_{jt} + X_{it} \beta + \theta \sum_{j \neq i} w_{ij} X_{jt} + \alpha_i + \eta_t + u_{it}, \quad (11)$$

$$u_{it} = \lambda \sum_{j \neq i} w_{ij} u_{jt} + \varepsilon_{it}, \quad (12)$$

where the dependent variable $Risk_{it}$ represents the level of debt default risk of city i in year t , which is endogenously determined by neighbouring cities' behaviours, $\sum_{j \neq i} w_{ij} Risk_{jt}$. Control variables X_{it} include regional economic and fiscal factors, as described in subsection 4.3. We use UCIBs' default risk calculated in subsection 4.1 as proxy for $Risk_{it}$. The spatial weight w_{ij} describes spatial interaction between city i 's and city j 's debt default risk. City fixed effect α_i and time fixed effect η_t capture region-level unobserved elements, such as variation in regional culture and endowment, and time-varying shocks respectively. ε_{it} is random disturbance term. The spatial autoregressive coefficient ρ , which captures the strength of spatial interactions among cities, is the main coefficient of interest.

The general expression of SPDM can be transformed into a set of simplified models by imposing restrictions on spatial autoregressive coefficient ρ , spatial autocorrelation coefficient λ , and the coefficient of spatial lag terms of explanatory variable θ . The simplified models include: the spatial Durbin model (SDM) if $\lambda = 0$; the spatial lag model (SLM) if $\lambda = 0$ and $\theta = 0$; the spatial error model (SEM) if $\rho = 0$ and $\theta = -\rho\beta$ (Anselin et al., 2008; Elhorst, 2014). We prefer SDM because it introduces spatially lagged explanatory variables to better specify the spatial interactions (Delgado et al., 2015). And it can also mitigate biases induced by possibly omitted or unobservable factors correlated with explanatory variables, and thus generate more unbiased estimation coefficients relative to SLM and SEM (Elhorst, 2012; LeSage & Pace, 2009). And we use the likelihood ratio test (LR) and Wald test to establish whether the spatial Durbin model could be simplified into SLM or SEM.

A great challenge to estimation of spatial interactions is the intrinsic endogenous problem. Because the dependent variable $Risk_{it}$ is simultaneously determined with spatial lags $\sum_{j \neq i} w_{ij} Risk_{jt}$, the ordinary least squares (OLS) estimates are biased and inconsistent (Cliff & Ord, 1973; Kelejian & Prucha, 1998). In addition, control variables may be endogenous. For example, the increasing debt default risk would slow down economic and fiscal growth. To tackle the endogeneity problem, we adopt maximum likelihood estimation (MLE) method in baseline estimation, and the generalized spatial two-stage least squares (GS2SLS) method in robustness check (Anselin, 1988; Brueckner, 2003). Economic or fiscal control variables are lagged one period to further alleviate potential endogeneity.

Another important issue in SDM is the spatial weight matrix $\mathbf{W}_n$, which defines relative political, economic or geographic distance among cities. In order to examine the possible explanations for spatial interactions, we construct five spatial weight matrices and all of them are restricted to be row-normalized with zeros on the diagonal, ensuring the weights lie between 0 and 1. We consider two spatial weight matrices to capture the strength of political competition among cities leaders, denoted by COMP1 and COMP2. Since city leaders in the same province are potential rivals in the Chinese political hierarchy, we construct COMP1 which takes city j as city i 's neighbour if they locate in the same province, to explore spatial interactions of debt default risk among political neighbours. However, with a decisive role of economic growth in RPE, political competition may only exist among prefectural-level cities with similar economic performance. For example, Jinan and Qingdao with similar GDP ranking in Shandong Province have intensified their economic competition for the status of 'National Central City'. But there is little competition between Qingdao with another city of Rizhao in the same province in the sense that the former ranks far higher than the latter. With the average *per capita* GDP of prefectural-level cities during our sample to measure economic performance, we construct COMP2 which treats city j as city i ' neighbour if these two cities are in the same province and city j 's *per capita* GDP lies within the three closest one to city i in the province. Thus, COMP2 detects spatial interactions among both political and economic neighbours.



We also build three across-province weight matrices to rule out alternative interpretations of spatial interactions. According to Tiebout (1956), Brueckner (2003) and Callander and Harstad (2015), economic spillover and resource flow are two main interpretations for spatial interactions besides political competition. Considering key channels of economic spillover and resource flow depend highly on geographic distance, we construct spatial matrices GEO1 and GEO2 to identify merely geographic neighbouring cities. GEO1 is a binary contiguity matrix, which defines city j as city i 's neighbour if city i and city j locate in different provinces but share a common border. And GEO2 takes those across-province cities whose geographical distance to city i is the closest three ones as its neighbours. We also explore the spatial interactions among merely economic neighbours to rule out the 'intellectual common trend' or policy effects (Manski, 1993). For example, the central government's debt regulation policy might lead to common trend in default risk for cities with similar economic conditions. So the spatial matrix FCOPM defines across-province cities whose *per capita* GDP is the three closest ones to city i as its neighbours.

4.3 | Data and variables

Due to local debts accumulated through UCIBs at relatively slow rate before 2010 as described in Section 2, the empirical analysis is conducted on balanced panel data of 245 prefectural-level cities from 2010 to 2017. Chinese treasury rate and UCIBs' basic information come from the WIND database, which records UCIBs' daily trading volume, daily opening and closing prices, time to maturity, issued amount and coupon rate. Three types of UCIBs are excluded, namely those issued by direct-controlled municipalities such as Beijing, Shanghai, Tianjin and Chongqing, those with maturity less than one year, and those with less than one month to maturity date. The final sample includes 5,084 UCIBs issued by 245 prefectural-level cities, most of which are issued by eastern cities. For example, Suzhou has issued 193 UCIBs accounting for 3.8% in the sample, whereas Shangluo only has issued three UCIBs. We calculate UCIBs' daily yield by using the difference between their daily opening price and closing price, and take an average of daily yields to generate quarterly yield. The UCIBs' credit ratings come from *Dagong Global Credit Rating* and *China Chengxin International Credit Rating*, the two authorized entities to assess bonds' credit ratings. And we assign 9 credit ratings of BBB, A-, A, A+, AA-, AA, AA+, AAA- and AAA to numbers from 0 to 8.

We control several economic and fiscal variables to reduce potential confounding factors. First, the pressure of economic growth is closely associated to local infrastructure projects, leading to variations in local debt accumulation and its default risk (Guo, 2009; Pan et al., 2017). We adopt one-period lagged *per capita* GDP (PGDP) and GDP growth rate (*Growth*) to measure this pressure, because a higher-level authority might impose strict sanctions on local leaders with worse economic achievement than previous ones. Second, increased fiscal pressure could drive local governments to resort to UCIBs, and thus bring a higher default risk. We use the one-period lagged ratio of gap between budgetary expenditure and revenue to revenue (*Gap*) to capture this element. Third, labour inflow reflects intense competition for high-quality labour (Davis & Haltiwanger, 2014; Heckman & Yi, 2012). This could broaden the tax base and decrease default risk on local government debt. Thus, we measure labour inflow using a one-period lagged ratio of net population inflow to cities' permanent population (*Inflow*). Fourth, as collateral for UCIBs, land leasing revenue plays an important role in paying local government debts and thereby reduces the default risk (Huang & Du, 2017; Pan et al., 2017). Therefore, we also control one-period lagged land leasing revenue (*Land*). Finally, with industrial structure upgrades, economic development relies more heavily on local infrastructure such as highways, commuter rails and the public telecommunications network. We characterize industrial structure by the one-period lagged ratio of secondary and tertiary industry production to GDP (*Indus*). While the land leasing data is extracted from *China Land and Resources Statistical Yearbook*, other control variables are collected from *China City Statistical Yearbook*. All the nominal variables are deflated with 2010 as base year. And except for *Amihud*, *Turnover*, *Coupon rate*, *Credit rating*, *Growth*, *Inflow*, *Gap* and *Indus*, other variables have been transformed to natural-log form for empirical estimation.



In subsection 5.4, we explore how spatial interactions are shaped by local officials' political incentives. We try to use city party secretary's age (*Age*) to capture incentives, and the age information is collated from Baidu Baike (<http://baike.baidu.com>).

Table 2 reports the summary statistics of main variables in the study. After subtracting the risk-free rate, the UCIBs' quarterly average yield spread is 55.90 bps. Average issued amount and coupon rate are roughly 1,000.2 billion RMB and 5.70%, respectively. The three indices used to measure UCIBs' liquidity, *Amihud*, *High-low*, and *Turnover*, have average values of 0.08%, 3.42 RMB, and 48.15%, respectively, implying fairly illiquid due to infrequent transactions of UCIBs. The bottom panel in Table 2 lists the economic characteristics at prefectural level. UCIBs' annual average default risks measured by *Default_Amihud*, *Default_High-low* and *Default_Turnover* are 631.22 bps, 627.81 bps and 515.54 bps, respectively.

5 | RESULTS

5.1 | Estimation of debt default risk

We use Equation (6) to examine whether UCIB's liquidity risk is correctly priced, and then calculate debt default risk according to Equation (10). As reported in Table 3, the coefficients of *Amihud*, *High-low* and *Turnover* are 6.6847,

TABLE 2 Summary statistics

Variable	Obs.	Mean	P10	P90	Std. dev
Characteristics of UCIBs:					
Yield_spread (Bps)	39,699	55.90	26.57	67.18	65.43
Amihud (%)	39,699	0.78	0.00	2.41	1.18
High_low (RMB)	39,699	3.42	0.82	6.84	4.71
Turnover (%)	39,699	48.55	0.00	114.44	134.76
Time to Maturity (year)	39,699	4.71	2.43	6.97	1.96
Issued amount (million)	5,084	1000.20	400.00	1800.00	625.57
Coupon rate (%)	5,084	5.70	3.99	7.29	1.22
Credit rating	5,084	5.71	5	8	1.12
Characteristics of prefectural-level cities:					
Default_Amihud (Bps)	1960	631.22	412.71	892.68	290.91
Default_Highlow (Bps)	1960	627.81	407.21	894.60	292.39
Default_Turnover (Bps)	1960	515.54	288.87	777.31	289.82
Default1	1960	0.09	0.11	0.00	0.29
Default2	1960	0.12	0.13	0.00	0.44
PGDP (thousand)	1960	53.19	19.75	94.52	14.84
Growth (%)	1960	9.98	5.95	14.91	4.72
Inflow (‰)	1960	3.73	−5.02	13.06	7.66
Gap (%)	1960	1.66	0.23	3.61	1.58
Indus (%)	1960	49.08	36.96	60.39	9.92
Land (billion RMB)	1960	10.37	0.91	24.77	17.62
Age (years)	1960	52.85	47.00	57.00	3.49



TABLE 3 Estimation of UCIB's liquidity risk

	Yield_spread (1)	Yield_spread (2)	Yield_spread (3)
Amihud	6.6847** (2.6103)		
High_low		3.3834*** (0.6765)	
Turnover			−0.0760*** (0.0233)
Time to maturity	−0.1822*** (0.0036)	−0.1809*** (0.0042)	−0.1782*** (0.0051)
Coupon rate	127.8007*** (2.5085)	126.0342*** (2.5403)	129.3453*** (2.4901)
Issued amount	8.3526*** (0.5208)	8.1908*** (0.5217)	8.3884*** (0.5212)
Credit rating	Y	Y	Y
Time fixed effect	Y	Y	Y
Obs.	39,699	39,699	39,699

Notes: Standard errors are in parenthesis. Statistical significance as follows.

***p < 0.01,

**p < 0.05,

*p < 0.1.

3.3834 and −0.0760 respectively, and significant at the 1% or 5% level. As discussed in subsection 4.1, the *Amihud* and *High-low* are positively correlated with liquidity risk, and *Turnover* is negatively correlated with liquidity risk. Thus, the results indicate that UCIB's liquidity risk is effectively embodied in its yield spread. The estimates for other control variables are mostly statistically significant with expected signs. For example, *Time to maturity* is negatively correlated with yield spread, while the coefficients of *coupon rate* and *issued amount* are significantly positive.

Using the estimated coefficients of *Amihud*, *High-low* and *Turnover*, we compute UCIB's default risk according to Equation (10). Figure 4 plots annual changes in UCIBs' average default risk between 2010 and 2017. Overall, the default risk remained stable before 2016, but it jumped abruptly afterwards because the State Council accepted little responsibility for the repayment of UCIBs. In addition, the trends of UCIBs' default risk are fairly consistent across the three measures, that is, *Default_amihud*, *Default_highlow* and *Default_turnover*, validating the calculation methods in subsection 4.1 to some extent.

Figure 5 summarizes the average annual change in UCIB's default risk across cities measured by *Default_amihud* between 2010 and 2017. We categorize cities into four groups by the quartiles of annual change rates in default risk. The average default risk increases by 97.5 Bps for the country as a whole from 2010 to 2017. From a regional perspective, cities from the top two quartiles concentrate on northern regions, especially Inner Mongolia and three northeast provinces, reflecting that the default risk accumulated relatively faster in these regions. A possible explanation is the dominance of state-owned enterprises (SOEs) in northern China. Compared with private enterprises, the SOEs rely more heavily on bank loans and fiscal support to finance capital investment (Dong & Putterman, 2003). This would squeeze out financial resources to repay matured government debts, thus leading to fast-growing default risk. Also, the declining total factor productivity and economic performance in northeast China could also lower debt servicing capacity (Wu et al., 2018).

5.2 | Baseline results

Table 4 reports the baseline results for spatial weight matrices COMP1 and COMP2. The LR test and Wald test are both significant at 1% or 5% level, indicating SDM is properly applied to describe spatial interactions of debt

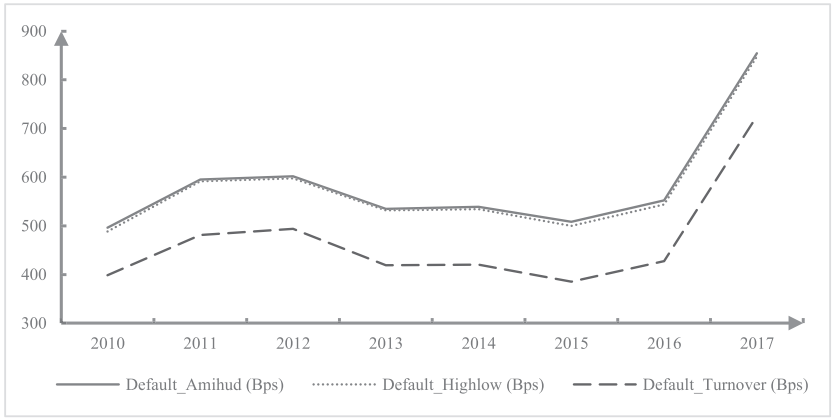


FIGURE 4 UCIBs' default risk across 2010–2017

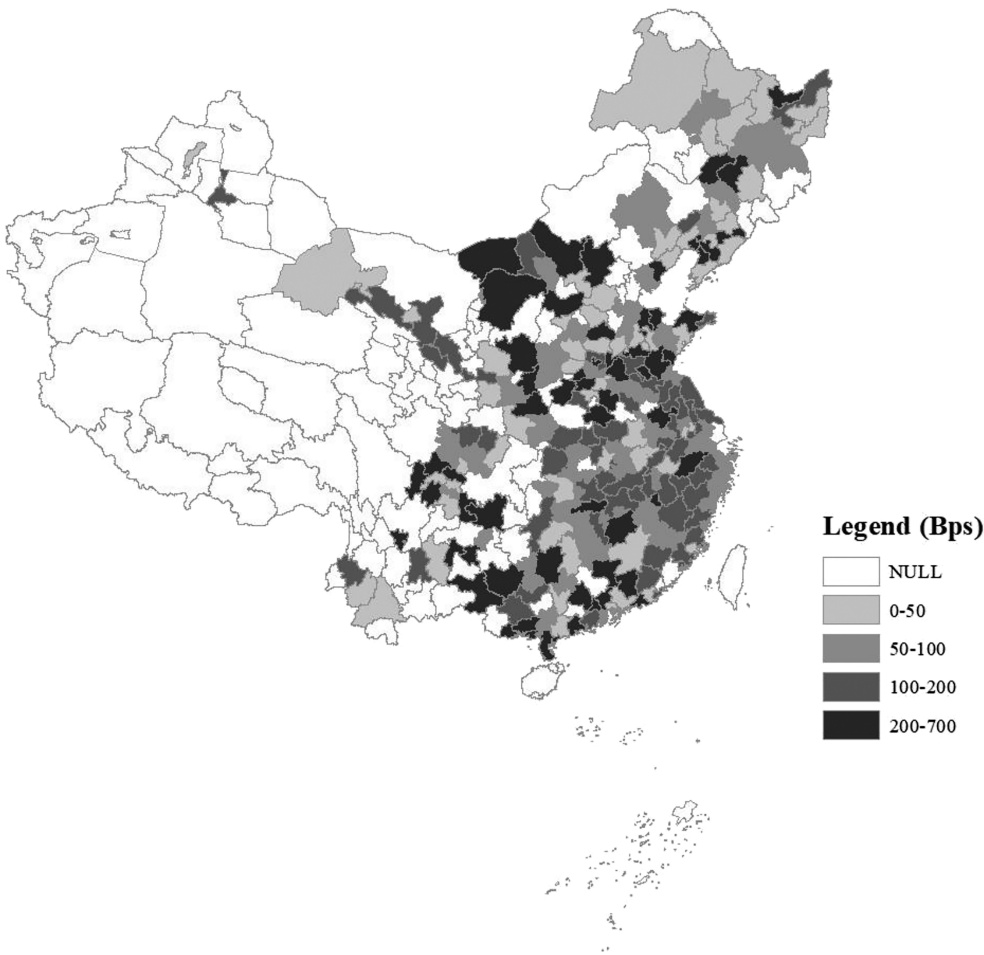


FIGURE 5 Change in UCIBs' default risk at prefectural level between 2010 and 2017



TABLE 4 Spatial interactions of default risk

	W: CAMP1			W: CAMP2		
	(1) Default_Amihud	(2) Default_Highlow	(3) Default_Turnover	(4) Default_Amihud	(5) Default_Highlow	(6) Default_Turnover
Wy	0.1911*** (0.0443)	0.1952*** (0.0441)	0.1293*** (0.0477)	0.2271*** (0.0249)	0.2277*** (0.0249)	0.2133*** (0.0253)
PGDP	0.0564* (0.0298)	0.0589* (0.0301)	0.0546 (0.0347)	0.0507* (0.0287)	0.0507* (0.0290)	0.0450 (0.0334)
Growth	0.0004 (0.0017)	0.0005 (0.0017)	0.0004 (0.0019)	0.0042*** (0.0016)	0.0045*** (0.0016)	0.0052*** (0.0018)
Inflow	−0.0019 (0.0017)	−0.0018 (0.0017)	−0.0023 (0.0020)	−0.0037** (0.0016)	−0.0038** (0.0016)	−0.0048** (0.0019)
Gap	0.0079 (0.0083)	0.0072 (0.0084)	0.0053 (0.0097)	0.0212** (0.0083)	0.0208** (0.0084)	0.0213*** (0.0096)
Indus	0.0978 (0.3830)	0.0616 (0.3869)	0.1191 (0.4460)	0.0254 (0.3750)	−0.0246 (0.3795)	0.0481 (0.4357)
Land	−0.0008 (0.0125)	0.0001 (0.0127)	−0.0017 (0.0146)	0.0067 (0.0116)	0.0084 (0.0118)	0.0098 (0.0135)
W*PGDP	0.0981 (0.0636)	0.1066* (0.0643)	0.1118 (0.0739)	−0.0146 (0.0344)	−0.0172 (0.0348)	−0.0334 (0.0399)
W*Growth	0.0056 (0.0036)	0.0058 (0.0036)	0.0075* (0.0042)	0.0114*** (0.0021)	0.0117*** (0.0021)	0.0145*** (0.0024)
W*Inflow	0.0011 (0.0027)	0.0012 (0.0027)	0.0020 (0.0031)	−0.0079*** (0.0020)	−0.0080*** (0.0020)	−0.0087*** (0.0023)
W*Gap	0.0829*** (0.0206)	0.0877*** (0.0209)	0.1105*** (0.0242)	0.0582*** (0.0127)	0.0584*** (0.0129)	0.0673*** (0.0147)
W*Indus	0.7877 (0.7067)	0.8620 (0.7141)	1.0727 (0.8236)	0.6490 (0.4872)	0.7141 (0.4928)	0.6724 (0.5662)
W*Land	−0.0247 (0.0259)	−0.0247 (0.0262)	−0.0210 (0.0301)	0.0165 (0.0156)	0.0171 (0.0157)	0.0229 (0.0181)
Spatial fixed effect	Y	Y	Y	Y	Y	Y
Time-period effect	Y	Y	Y	Y	Y	Y
Wald test spatial lag	19.11***	20.84***	24.08***	15.31**	16.52***	16.67**
LR test spatial lag	19.27***	21.03***	24.32***	15.28**	16.64**	16.62**
Wald test spatial error	21.75***	23.71***	25.93***	17.03***	18.35***	17.76***
LR test spatial error	22.19***	24.24***	26.61***	18.72***	19.45***	19.15***
Hausman test	0.0016	0.0014	0.0015	0.0016	0.0021	0.0000

Notes: Standard errors clustered at the city level are reported in parentheses. Statistical significance as follows:

***p < 0.01,
**p < 0.05,
*p < 0.1.



default risk. And the Hausman test results support fixed effects specification. Columns (1)–(3) present the estimation results for COMP1, with political neighbours defined as all the prefectural-level cities in the same province. With default risk measured by *Default_amihud*, *Default_highlow* and *Default_turnover*, the coefficients for spatially lagged dependent variables (*Wy*) are about 0.1911, 0.1952 and 0.1293 respectively, and remain significant at the 1% level. The results imply that local leaders would mimic political rivals' behaviour in economic development and risk reduction, thus resulting in spatial interactions among prefectural-level cities in terms of debt default risk.

Using COMP2, we further examine the spatial interactions of default risk among cities sharing parent province and similar economic conditions. The coefficients in columns (4)–(6) are larger than those in columns (1)–(3), implying local governments mimic and react more strongly to direct contestants defined by COMP2. This finding is consistent with Yu et al. (2016), which documents stronger spatial interactions of local government investment between cities with similar rankings of *per capita* GDP in the same province.

The political competition seems to be fundamental reason behind the spatial interactions of debt default risk. To stand out in competition, local leaders in the same province issue UCIBs and finance economic development strategically to impress their superiors, leading to spatial correlations in local government debts and its default risk. Especially, cities leaders with similar economic achievements are more likely to be political rivals (Li et al., 2019; Ma, 2016). Hence, local leaders react more intensively to policy changes of both political and economic neighbours. On the other hand, increased weight of risk reduction in RPE urges local leaders to reduce the default risk of government debt strategically to avoid lagging behind their political rivals. And for cities with similar economic achievement, the risk reduction would become more critical in political competition. Therefore, the political and economic neighbouring cities will experience even stronger spatial interactions of default risk than mere political neighbours.

Regarding the control variable and its spatial lag, the coefficients in Table 4 cannot evaluate marginal impacts due to spatial autocorrelation in SDM (Elhorst, 2014). Therefore, following LeSage and Pace (2009), we decompose the impacts of control variables into direct and indirect effects, which respectively represent the

TABLE 5 Decomposition of effects on default risk: direct and indirect effects

	Default_Amihud		Default_Highlow		Default_Turnover	
	(1) Direct effects	(2) Indirect effects	(3) Direct effects	(4) Indirect effects	(5) Direct effects	(6) Indirect effects
PGDP	0.0515* (0.0288)	−0.0047 (0.0396)	0.0513* (0.0292)	−0.0079 (0.0401)	0.0448 (0.0335)	−0.0300 (0.0456)
Growth	0.0048*** (0.0015)	0.0151*** (0.0024)	0.0052*** (0.0015)	0.0157*** (0.0024)	0.0060*** (0.0017)	0.0190*** (0.0027)
Inflow	−0.0041*** (0.0015)	−0.0107*** (0.0021)	−0.0041*** (0.0015)	−0.0109*** (0.0021)	−0.0052*** (0.0018)	−0.0118*** (0.0024)
Gap	0.0249*** (0.0080)	0.0785*** (0.0146)	0.0246*** (0.0081)	0.0787*** (0.0148)	0.0253*** (0.0093)	0.0882*** (0.0168)
Indus	0.0643 (0.3525)	0.7957 (0.5491)	0.0176 (0.3567)	0.862 7 (0.5558)	0.0856 (0.4098)	0.8163 (0.6310)
Land	0.0084 (0.0114)	0.0233 (0.0185)	0.0102 (0.0115)	0.0244 (0.0187)	0.0119 (0.0132)	0.0315 (0.0212)

Notes: Standard errors clustered at the city level are reported in parentheses. Statistical significance as follows:

***p < 0.01,

**p < 0.05,

*p < 0.1.



average impacts of change in control variables of a particular city on the default risk of its own and neighbouring cities. Table 5 reports direct and indirect effects based on SDM, with spatial matrix COMP2. With respect to the direct effects in columns (1), (3) and (5), the coefficients of economic pressure (*PGDP* and *Growth*) are significantly positive. This implies that economic pressure prompts local leaders to issue more UCIBs to finance government investment and economic growth, resulting in a surge of debt default risk. The labour inflow (*Inflow*) contributes to lower debt default risk through improvement of labours' quality and economic efficiency. And fiscal pressure (*Gap*) has a significantly positive impact on debt default risk due to poor debt repayment capability. However, the coefficients of land leasing fees (*Land*) are insignificant, which seems to deviate from the expectation at a first glance. Undoubtedly, excessive land leasing fees from Chinese booming estate market improve debt serving ability, namely debt-repaying effects. But it also provides local governments better collateral to borrow more from financial markets (land-financing effects), which could translate into higher debt volume and default risk (Zhang et al., 2018). Thus debt-repaying effects and land-financing effects might cancel each other out in some way, resulting in insignificant coefficient of *Land*. As for the indirect effects in columns (2), (4) and (6), the pressure of economic growth leads to higher debt default risk in neighbouring cities. It indicates that one city's economic achievements would prompt their neighbours to set higher economic growth targets, thus leading to rising default risk. And the *Inflow* and *Gap* also affect neighbouring cities' debt default risk through spill-over channels.

In the Appendix, we also adopted the GS2SLS method, and then alternate measurement of default risk with *Default1* and *Default2* to check the robustness of empirical results. The coefficients of spatial lags remain similar to those in Table 4, thus indicating significant and robust spatial interactions of debt default risk.

5.3 | Promotion incentives

In support of political competition, we further examine the impacts of local leaders' personal characteristics on spatial interactions of default risk. According to Alesina et al. (2019), local leaders' age is an important factor affecting political incentives. Under China's mandatory retirement regulation, it is common practice to transfer city leaders to semi-retirement positions one or two years before their retirement age of 60 (Kou & Tsai, 2014). This arrangement means city leaders might experience the upsurge of incentives to engage in political competition several years before approaching the age limit of promotion (Shi & Xi, 2018). We choose the 54–58 age range, the last possible period for city leaders to be promoted, and construct age dummy variable to capture the effects of political incentives in shaping city officials' strategic behaviours.

Table 6 presents the estimates using MLE. The coefficients of spatial lag are still significantly positive in all columns. And for the interaction term between age dummy variable and spatial lag, the coefficients are significantly positive at 1% or 5% levels, except in column (2). This suggests that the spatial interactions of default risk vary with city leaders' ages. For city party secretaries aged 54 to 58, it is urgent to surpass their political rivals in RPE to catch the last chance for political promotion before the age limit. Therefore, they tend to respond more intensely to policy changes in other jurisdictions, contributing to stronger spatial interactions of default risk.

5.4 | Alternative explanations

The current results lend supports to the political competition in shaping spatial interactions of debt default risk. However, there might be other alternative explanations, for example, the economic spillover and resource flow hypotheses. Given that the economic spillovers and resource flows are more likely to appear among geographical neighbouring cities, we construct binary contiguity matrix GEO1 and geographic proximity matrix GEO2 to examine these hypotheses. Results based on GEO1 are reported in columns (1)–(3) in Table 7. The magnitude of spatial lags'



TABLE 6 The political incentives and spatial interactions of default risk

	W: CAMP1			W: CAMP2		
	(1) Default_ Amihud	(2) Default_ Highlow	(3) Default_ Turnover	(4) Default_ Amihud	(5) Default_ Highlow	(6) Default _Turnover
Wy	0.1716*** (0.0454)	0.1732*** (0.0453)	0.1271*** (0.0478)	0.1766*** (0.0258)	0.1801*** (0.0258)	0.1668*** (0.0261)
1(54 ≤ Age ≤ 58)	−0.0412* (0.1527)	−0.0467** (0.1512)	−0.0453* (0.1521)	−0.0506 (0.1494)	−0.0505** (0.1478)	−0.0490* (0.1483)
Wy*1(54 ≤ Age ≤ 58)	0.1915** (0.0804)	0.1919** (0.0791)	0.1242 (0.0805)	0.1943*** (0.0445)	0.1901*** (0.0440)	0.1761*** (0.0452)
Spatial fixed effect	Y	Y	Y	Y	Y	Y
Time-period effect	Y	Y	Y	Y	Y	Y
Wald test spatial lag	17.24***	18.59***	19.87***	14.11**	15.05**	16.39**
LR test spatial lag	18.94**	20.24***	21.62***	19.54**	20.78***	22.71***
Wald test spatial error	17.49***	18.80***	20.25***	14.75**	15.70**	17.10***
LR test spatial error	26.34***	28.11***	26.37***	23.45***	24.50***	24.22***
Hausman test	0.0020	0.0034	0.0025	0.0015	0.0017	0.0010

Notes: Standard errors clustered at the city level are reported in parentheses. Statistical significance as follows:

***p < 0.01,

**p < 0.05,

*p < 0.1.

coefficients becomes relatively small and insignificant, indicating little spatial interactions among geographically proximate cities in different provinces. And this conclusion is also supported by estimates using GEO2 in columns (4)–(6). Thus, these results reveal that the economic spillover and resource flow fail to drive the spatial interaction in default risk, which support the political competition hypothesis.

We also examine the spatial interactions among merely economic neighbours located in different provinces as a placebo test in the sense that little political competition appears among these local leaders according to China's political hierarchy. The results in Table 8 indicate these cities do not respond to change in default risk of their merely economic neighbours, with coefficients of spatial lags close to zero. This further confirms that the spatial interactions are not driven by 'intellectual common trend' or policy effects among cities with similar economic potential. Taken together, the spatial interaction in debt default risk reflects political competition among local leaders, instead of other factors.

5.5 | Heterogeneity in spatial interactions of default risk

As discussed in Section 2 and subsection 5.1, there is uneven geographic distribution of local government debts and its default risk. Although eastern cities witnessed massively increased debt in the last decade, their higher land price and land revenue enhance debt repayment capability and thus alleviate the default risk. Meanwhile, local government debts and its default risk in central-western cities grew rapidly in recent years due to a host of infrastructure projects. In order to explore heterogeneity in spatial interactions of debt default risk across regions, we divide the sample into two subsamples of the eastern, and the central-western, and then repeat the baseline estimation using COMP2.



TABLE 7 Spatial interactions among merely geographic neighbours

	W: GEO1			W: GEO2		
	(1) Default_Amihud	(2) Default_Highlow	(3) Default_Turnover	(4) Default_Amihud	(5) Default_Highlow	(6) Default_Turnover
Wy	0.0425 (0.0379)	0.0381 (0.0377)	0.0578 (0.0372)	−0.0116 (0.0218)	−0.0106 (0.0216)	−0.0164 (0.0222)
Spatial fixed effect	Y	Y	Y	Y	Y	Y
Time-period effect	Y	Y	Y	Y	Y	Y
Wald test spatial lag	18.55***	17.63***	19.31***	18.72***	17.56***	23.17***
LR test spatial lag	25.26***	24.26***	23.07***	18.63***	17.49***	23.08***
Wald test spatial error	17.75***	17.14***	20.38***	18.57***	17.45***	24.91***
LR test spatial error	26.15***	25.57***	22.67***	18.45***	17.35***	22.74***
Hausman test	0.0001	0.0001	0.0017	0.0000	0.0000	0.0000

Notes: Standard errors clustered at the city level are reported in parentheses. Statistical significance as follows:

***p < 0.01,
**p < 0.05,
*p < 0.1.



TABLE 8 Spatial interactions among merely economic neighbours

	(1) Default_Amihud	(2) Default_Highlow	(3) Default_Turnover
Wy	0.0179 (0.0178)	0.0129 (0.0177)	0.0206 (0.0178)
Spatial fixed effect	Y	Y	Y
Time-period effect	Y	Y	Y
Wald test spatial lag	23.93***	22.48***	19.61***
LR test spatial lag	23.78***	22.35***	19.52***
Wald test spatial error	22.76***	21.24***	18.94***
LR test spatial error	22.63***	21.13***	18.85***
Hausman test	0.0000	0.0000	0.0000

Notes: Standard errors clustered at the city level are reported in parentheses. Statistical significance as follows:

***p < 0.01,

**p < 0.05,

*p < 0.1.

Table 9 reports the estimated results of eastern cities, and central-western cities. All coefficients of spatial lags are still significantly positive, implying spatial interactions induced by political competition are pervasive across different regions. And it is worth noting that coefficients in central-western cities in columns (4)–(6) are much larger. Although central-western cities have relatively small-scale government debts, their lower fiscal revenue and poor risk management compared to the eastern region would result in less frequent transactions and wildly fluctuating trading prices of UCIBs, in the sense that market traders would decide their transaction behaviours based on local fundamentals. Thus, some central-western cities have experienced high default risk. Another possible explanation is that the political competition among central-western cities is more intense than eastern cities. Thus, local officials in these regions put more effort into economic development and risk management to get political promotion, resulting in stronger spatial interactions. Furthermore, the spatial interactions of debt default risk may pose the synchronization problem to the financial stability, in the sense that debt default in some cities will spread to other cities and possibly trigger the so called ‘Minsky moment’.

5.6 | Policy change in local financing by the new Budget Law

With concerns of the opaqueness in the local government debts, the State Council passed the new Budget Law in 2015, which allows prefectural-level cities to issue local government bonds to finance infrastructure projects and public services. The permission for direct market borrowing not only lowers cities’ financing reliance on UCIBs, but improves transparency of local government debts, which presumably reduce the spatial interactions of debt default risk. We separate sample intervals into two sub-intervals of 2010–2015 and 2016–2017, and then compare the coefficients of spatial lags in different subsamples.

The estimated results are shown in Table 10. Compared to the sub-interval 2010–2015, the coefficients of spatial lags after the policy in columns (4)–(6) are relatively smaller. According to the new Budget Law, local governments are allowed to resort to direct and transparent market borrowing to finance infrastructure and economic development. It virtually helps higher-level authorities to supervise and regulate local debt, thus lowers the spatial interactions of debt default risk. However, due to unchanged political promotion among local leaders, the spatial interactions of debt default risk are not fully eliminated.



TABLE 9 Heterogeneity in spatial interaction across the eastern and central-western cities

	Eastern cities			Central-Western cities		
	(1) Default_Amihud	(2) Default_Highlow	(3) Default_Turnover	(4) Default_Amihud	(5) Default_Highlow	(6) Default_Turnover
Wy	0.1371*** (0.0441)	0.1450*** (0.0439)	0.1225*** (0.0447)	0.2189*** (0.0316)	0.2143*** (0.0318)	0.2093*** (0.0321)
Spatial fixed effect	Y	Y	Y	Y	Y	Y
Time-period effect	Y	Y	Y	Y	Y	Y
Wald test spatial lag	21.33**	21.58***	18.23***	19.39**	20.31***	21.38**
LR test spatial lag	21.22**	21.47***	17.58**	19.03***	20.20***	21.25**
Wald test spatial error	20.72***	20.87***	18.07**	21.06***	22.02***	22.64***
LR test spatial error	20.46**	20.31***	17.21*	21.00***	21.96***	22.58***
Hausman test	0.0271	0.0103	0.0127	0.0451	0.0325	0.0218

Notes: Standard errors clustered at the city level are reported in parentheses. Statistical significance as follows:

***p < 0.01,
**p < 0.05,
*p < 0.1.



TABLE 10 Effects of policy change in local financing

	2010–2015			2016–2017		
	(1) Default_Amihud	(2) Default_Highlow	(3) Default_Turnover	(4) Default_Amihud	(5) Default_Highlow	(6) Default_Turnover
Wy	0.2189*** (0.0292)	0.2216*** (0.0291)	0.2077*** (0.0297)	0.1270* (0.0541)	0.1327* (0.0538)	0.1142* (0.0550)
Spatial fixed effect	Y	Y	Y	Y	Y	Y
Time-period effect	Y	Y	Y	Y	Y	Y
Wald test spatial lag	18.13**	17.62**	17.11**	16.97***	16.38**	15.74**
LR test spatial lag	17.42**	17.29**	16.55**	16.68**	16.12**	15.19**
Wald test spatial error	16.21***	15.89***	15.07**	15.21***	14.63**	14.71**
LR test spatial error	17.31**	16.54**	15.89**	15.69**	15.13**	15.32**
Hausman test	0.0000	0.0003	0.0001	0.0001	0.0001	0.0000

Notes: Standard errors clustered at the city level are reported in parentheses. Statistical significance as follows:

***p < 0.01,

**p < 0.05,

*p < 0.1.



6 | CONCLUSIONS

This paper explores the spatial interactions of default risk on local government debts. We first propose a theoretical framework to explain the mechanism behind strategic interactions among local leaders, and check the hypotheses using balanced panel data of 245 prefectural-level cities from 2010 to 2017. The empirical results indicate positive spatial interactions of default risk among political neighbouring cities, but little spatial interactions emerge among merely geographic or economic neighbours. And further evidence implies that city leaders' political incentives drive up the spatial interactions. Taken together, the empirical analysis suggests that political competition is the fundamental reason for spatial interactions of default risk. We also examine the effects of policy change in local financing due to the new Budget Law in 2015. The spatial interaction of default risk reduced to some extent, but is not fully removed after local governments gain access to direct market borrowing authorized by the law.

Based on this empirical evidence, we argue that while performance-based political competition boosts miraculous economic growth in China, it also forms a debt-driven economy and brings about spatial interactions of debt default risk. The policy implications are three-fold. First, it is important to further clarify the local governments' responsibility for UCIBs' repayment, and confirm that central government provides no explicit or implicit credit guarantee for local government debt. Moreover, the regulators should pay more attention to the spatial patterns in default risk. Especially, risk management should be strengthened to prevent the Minsky moment of risk eruption at the same time. Second, it is urgent to provide local governments more direct and transparent financing vehicles, and allow local governments more autonomy in local government bonds to lower spatial interactions of debt default risk. Finally, it is necessary to assign more weights to environmental protection, risk prevention and public services in RPE so as to change the dominant role of economic indicators in political promotion. Actually, the central government has gradually adjusted evaluation targets since the 13th five-year plan to solve socio-economic problems arising from previous GDP-based tournament.

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APPENDIX A

In this section, we adopt the GS2SLS method and alternate measures of debt default risk to assess robustness of empirical results. The estimated results using GS2SLS are presented in Table A1. Following Kelejian and Prucha (1998) and Drukker et al. (2013), we regress spatially lagged dependent variable Wy on a set of instruments $H = [X, WX, W^2X]$ in the first stage estimation, and then use the fitted value to estimate autoregressive parameter ρ with generalized method of moments. The Hausman's endogeneity test in Table A1 indicates the spatial lags are

TABLE A1 Robustness check: change of estimation method

	W: CAMP1			W: CAMP2		
	(1) Default_ Amihud	(2) Default_ Highlow	(3) Default_ Turnover	(4) Default_ Amihud	(5) Default_ Highlow	(6) Default_ _Turnover
Wy	0.2503*** (0.0793)	0.2677*** (0.0787)	0.2952*** (0.0876)	0.2897*** (0.0286)	0.2910*** (0.0281)	0.2965*** (0.0291)
Year dummy	Y	Y	Y	Y	Y	Y
First stage F test	[0.0020]	[0.0002]	[0.0006]	[0.0000]	[0.0000]	[0.0000]
Sargan's overidentification test	[0.0006]	[0.0002]	[0.0005]	[0.0013]	[0.0004]	[0.0002]
Hausman's endogeneity test	[0.0002]	[0.0000]	[0.0000]	[0.0006]	[0.0002]	[0.0000]

Notes: Standard errors clustered at the city level are reported in parentheses. Statistical significance as follows:

***p < 0.01,

**p < 0.05,

*p < 0.1.



endogenous, and the chosen instruments H satisfy the relevance condition with first stage F test significant at 1% level. Evidently, the coefficients of spatial lags remain similar to baseline results in Table 4, confirming validity of baseline results.

Furthermore, we use *Default1* and *Default2* to measure debt default risk and repeat the baseline estimation. As reported in Table A2, the coefficients of Wy are all significantly positive, and coefficients in column (3) and (4) using COMP2 are much larger than those in column (1) and (2), which is consistent with Table 4. To sum up, the empirical results prove significant and robust spatial interactions of debt default risk among political neighbouring cities.

TABLE A2 Robustness check: change of default risk measurement

	COMP1		COMP2	
	(1) Default1	(2) Default2	(3) Default1	(4) Default2
Wy	0.2001*** (0.0278)	0.1849*** (0.0266)	0.3140*** (0.0376)	0.2861*** (0.0392)
Spatial fixed effect	Y	Y	Y	Y
Time-period effect	Y	Y	Y	Y
Wald test spatial lag	36.37***	36.28***	18.94***	18.78***
LR test spatial lag	36.19***	39.54***	19.03***	21.00***
Wald test spatial error	61.45***	69.58***	22.53***	22.86***
LR test spatial error	62.97***	65.84***	23.34***	26.45***
Hausman test	0.0006	0.0229	0.0000	0.0039

Notes: Standard errors clustered at the city level are reported in parentheses. Statistical significance as follows:

- ***p < 0.01,
- **p < 0.05,
- *p < 0.1.



Resumen. El artículo investiga el mecanismo institucional que está detrás del aumento de la deuda de los gobiernos locales y del riesgo de impago en la última década en China. Utilizando datos a nivel de prefectura de 2010 a 2017, se estudió el riesgo de impago de la deuda de los gobiernos locales a través de la lente de las interacciones espaciales. Los resultados empíricos identifican interacciones espaciales positivas del riesgo de impago entre ciudades vecinas en lo político, pero no para las vecinas meramente en lo geográfico o lo económico. Además, las interacciones espaciales varían en función de los incentivos políticos de los líderes locales. Estos resultados sugieren en conjunto que las interacciones espaciales del riesgo de impago están impulsadas por la competencia política entre los líderes locales. El estudio también descarta explicaciones alternativas, como los efectos del flujo de recursos.

抄録: 本稿では、過去10年間で中国における増加する地方政府債務と債務不履行リスクの背後にある制度的メカニズムを検討する。2010~2017年の県レベルのデータを用いて、地方自治体の債務不履行リスクを空間的相互作用の観点から分析する。実証的分析から、単に地理的や経済的に近いだけでなく、政治的に近い都市間において債務不履行リスクの正の空間的相互作用があることが確認される。さらに、この空間的相互作用は地方指導者の政治的動機に左右される。以上の知見から、債務不履行リスクの空間的相互作用が地域の指導者間の政治的競争によって引き起こされていることが示唆される。また、この研究の結果から、資源フロー効果などのその他の仮説は否定される。